

## KCS

KCS stands for Knowledge-Centered Service, a methodology that aims to improve service delivery and knowledge management in service organizations. Some of the main features and benefits of KCS are :

- KCS is based on the principle of “capture, structure, reuse, and improve” knowledge as part of the service process.
- KCS enables service agents to create and update knowledge articles in real time, based on the issues and solutions they encounter during service interactions.
- KCS empowers service agents to search and access the existing knowledge base before escalating or creating new tickets, reducing duplication and increasing efficiency.
- KCS fosters a culture of collaboration and feedback among service agents, who can rate, comment, and flag knowledge articles for quality and relevance.
- KCS improves customer satisfaction and loyalty by providing consistent, accurate, and timely solutions to their problems.
- KCS reduces costs and increases productivity by minimizing rework, increasing first-contact resolution, and freeing up service agents for more complex and value-added tasks.

KCS is not to be confused with KCS, the abbreviation for Kansas City Southern, a railroad company that operates in the United States and Mexico .

## **Unit 1 - Brief history of microbiology, Types of microorganisms, Basic idea of domain bacteria, proteobacteria, non proteobacteria Gram –ve and Gram +ve bacteria, lichens, algae, protozoa, helminthes, viral structures, viral multiplication, Role of microorganisms in the production of industrial chemicals and pharmaceuticals**

- **Brief history of microbiology**
  - Microbiology is the scientific study of microorganisms, which are small or minute living things that are usually invisible to the naked eye.
  - Microbiology began with the invention of the microscope in the 16th century. Some of the pioneers of microbiology are:
    - \* Robert Hooke (1635-1703), an English scientist who observed fungi and plant cells under a microscope and coined the term “cell” .
    - \* Antonie van Leeuwenhoek (1632-1723), a Dutch draper and amateur microscopist who discovered bacteria, protozoa, and other microorganisms in various samples of water, saliva, and dental plaque .

- \* Ferdinand Cohn (1828-1898), a German botanist and microbiologist who classified bacteria into four groups based on their shape and founded the field of bacteriology.
- Microbiology also developed with the advancement of various techniques and theories, such as:
  - \* Staining, which is the use of dyes to enhance the contrast and visibility of microorganisms under a microscope.
  - \* Culture, which is the growth of microorganisms in artificial media under controlled conditions.
  - \* Koch's postulates, which are a set of criteria to establish the causal relationship between a microbe and a disease.
  - \* Pasteurization, which is the heating of liquids to kill harmful microorganisms and prevent spoilage.
  - \* Immunology, which is the study of the immune system and its response to pathogens.
  - \* Germ theory, which is the idea that many diseases are caused by microorganisms.
  - \* Antibiotics, which are substances that can kill or inhibit the growth of bacteria.
  - \* Biotechnology, which is the use of microorganisms or their products for various applications in medicine, agriculture, industry, and environment.
- **Types of microorganisms**
  - Microorganisms are diverse and can be classified into different groups based on their structure, function, and phylogeny. Some of the major types of microorganisms are:
    - \* Bacteria, which are prokaryotic cells that lack a nucleus and have a cell wall made of peptidoglycan. They can be found in almost every habitat and can perform various metabolic processes, such as photosynthesis, respiration, fermentation, and nitrogen fixation.
    - \* Archaea, which are prokaryotic cells that lack a nucleus and have a cell wall made of different substances than bacteria. They are often adapted to extreme environments, such as high temperature, salinity, acidity, or alkalinity. They can also perform various metabolic processes, such as methanogenesis, sulfate reduction, and halorespiration.
    - \* Eukarya, which are eukaryotic cells that have a nucleus and other membrane-bound organelles. They include four kingdoms of microorganisms: protists, fungi, plants, and animals.
      - Protists, which are mostly unicellular eukaryotes that can be autotrophic or heterotrophic. They include algae, protozoa, slime molds, and water molds.
      - Fungi, which are multicellular or unicellular eukaryotes that are heterotrophic and decompose organic matter. They include molds, yeasts, mushrooms, and lichens.

- Plants, which are multicellular eukaryotes that are autotrophic and use photosynthesis to make organic molecules. They include algae, mosses, ferns, and flowering plants.
- Animals, which are multicellular eukaryotes that are heterotrophic and ingest organic matter. They include sponges, worms, insects, and vertebrates.
- \* Viruses, which are acellular entities that consist of a nucleic acid (DNA or RNA) surrounded by a protein coat. They are obligate intracellular parasites that can infect all types of cells and cause various diseases.

### **Brief history of microbiology**

- Microbiology is the scientific study of microorganisms, which are usually too small to be seen with the naked eye.
- Microbiology has its origins in the 16th century, when some people observed the existence of tiny creatures in various substances, such as vinegar and milk.
- The invention of the microscope in the 17th century enabled the systematic observation and documentation of microorganisms.
- Robert Hooke (1635-1703) was an English scientist who used a microscope to examine cork and other plant materials. He coined the term “cell” to describe the basic unit of life.
- Anton van Leeuwenhoek (1632-1723) was a Dutch draper and amateur microscopist who made his own lenses and microscopes. He was the first to observe and describe bacteria, protozoa, fungi, and other microorganisms, which he called “animalcules”.
- In the 18th and 19th centuries, microbiology developed as a branch of biology and medicine, with contributions from many scientists, such as Louis Pasteur, Robert Koch, Joseph Lister, and Alexander Fleming.
- They discovered the roles of microorganisms in fermentation, disease, immunity, and antibiotics, and developed methods to isolate, culture, and identify them.
- In the 20th and 21st centuries, microbiology expanded to include the study of viruses, molecular biology, genetics, biotechnology, and ecology of microorganisms.
- Microbiology has applications in various fields, such as agriculture, industry, health, and environment.

### **Types of microorganisms**

- Microorganisms or microbes are microscopic organisms that exist as unicellular, multicellular, or cell clusters .
- Microorganisms are widespread in nature and are beneficial to life, but some can cause serious harm.
- Microorganisms can be divided into six major types: bacteria, archaea,

fungi, protozoa, algae, and viruses .

- Each type has a characteristic cellular composition, morphology, mean of locomotion, and reproduction.

### **Bacteria**

- Bacteria are prokaryotic cells, meaning that they do not have a nucleus or other membrane-bound organelles .
- Bacteria are the most abundant and diverse group of microorganisms, found in almost every environment on Earth .
- Bacteria can be classified by their shape, cell wall structure, metabolic activity, genetic makeup, and response to oxygen .
- Bacteria can cause diseases such as tuberculosis, cholera, syphilis, and anthrax, but they can also be beneficial for processes such as fermentation, decomposition, nitrogen fixation, and bioremediation .

### **Archaea**

- Archaea are also prokaryotic cells, but they differ from bacteria in their genetic, biochemical, and structural features .
- Archaea are often found in extreme environments, such as hot springs, salt lakes, deep sea vents, and acidic soils, where they are called extremophiles .
- Archaea are not known to cause any human diseases, but they play important roles in the global carbon and nitrogen cycles .

### **Fungi**

- Fungi are eukaryotic cells, meaning that they have a nucleus and other membrane-bound organelles .
- Fungi are heterotrophic, meaning that they obtain their energy from organic molecules .
- Fungi can be unicellular, such as yeasts, or multicellular, such as molds and mushrooms .
- Fungi can cause diseases such as ringworm, athlete's foot, candidiasis, and histoplasmosis, but they can also be beneficial for decomposing organic matter, producing antibiotics, and fermenting foods and beverages .

### **Protozoa**

- Protozoa are eukaryotic cells that are usually unicellular and motile .
- Protozoa can be classified by their mode of locomotion, such as flagella, cilia, or pseudopodia .
- Protozoa can be free-living or parasitic, meaning that they can live independently or depend on a host organism for survival .
- Protozoa can cause diseases such as malaria, amoebic dysentery, giardiasis, and sleeping sickness, but they can also be beneficial for grazing on

bacteria and algae, recycling nutrients, and serving as food sources for other organisms .

### **Algae**

- Algae are eukaryotic cells that are usually photosynthetic, meaning that they can convert light energy into chemical energy .
- Algae can be unicellular, such as diatoms and dinoflagellates, or multicellular, such as seaweeds and kelps .
- Algae can be found in aquatic and moist environments, where they form the base of many food chains .
- Algae can cause problems such as algal blooms, red tides, and water pollution, but they can also be beneficial for producing oxygen, biofuels, and food supplements .

### **Viruses**

- Viruses are acellular, meaning that they are not composed of cells .
- Viruses are obligate intracellular parasites, meaning that they can only replicate inside a living host cell .
- Viruses consist of a nucleic acid (DNA or RNA) surrounded by a protein coat, and sometimes a lipid envelope .
- Viruses can infect all types

### **Basic idea of domain bacteria**

- Domain bacteria is one of the three domains of life, along with domain archaea and domain eukarya.
- Domain bacteria includes the prokaryotes people encounter on an everyday basis. Prokaryotes are organisms that lack a membrane-bound nucleus and other organelles.
- Most bacterial species are heterotrophic; that is, they acquire their food from organic matter. The largest number of bacteria are saprobic, meaning that they feed on dead or decaying organic matter. A few bacterial species are parasitic; these bacteria live within host organisms and cause disease.
- Bacteria can also be classified by their shape, cell wall structure, metabolic pathways, and response to oxygen and other environmental factors.
- The domain bacteria is divided into several phyla, such as proteobacteria, cyanobacteria, firmicutes, actinobacteria, and spirochaetes. Each phylum contains different groups of bacteria with distinct characteristics and functions.
- Bacteria play important roles in the production of industrial chemicals and pharmaceuticals, such as antibiotics, enzymes, vitamins, hormones, and biofuels. Bacteria also participate in various biogeochemical cycles, such as the nitrogen cycle, the carbon cycle, and the sulfur cycle.
- Bacteria are essential for the health and survival of many living organisms, including humans. Bacteria help in digestion, immunity, and synthesis of

vitamins and other nutrients. Bacteria also form symbiotic relationships with other organisms, such as lichens, algae, protozoa, and helminthes.

### **Proteobacteria**

- Proteobacteria is a phylum of gram-negative bacteria discovered by Carl Woese in the 1980s based on nucleotide sequence homology .
- Proteobacteria are further classified into the classes alpha-, beta-, gamma-, delta- and epsilonproteobacteria, each class having separate orders, families, genera, and species.
- Proteobacteria include a wide variety of pathogenic and free-living bacteria, such as *Escherichia*, *Salmonella*, *Vibrio*, *Yersinia*, *Legionella*, and many others.
- Proteobacteria are also involved in nitrogen fixation, carbon monoxide oxidation, and sulfur cycling .
- Proteobacteria have diverse shapes, metabolisms, and habitats, ranging from soil and water to animal and plant hosts .

Some examples of proteobacteria are:

- Alphaproteobacteria: oligotrophs, organisms capable of living in low-nutrient environments such as deep oceanic sediments, glacial ice, or deep undersurface soil. Some examples are *Rhizobium* (nitrogen-fixing symbionts of legumes), *Rickettsia* (intracellular parasites of animals), and *Caulobacter* (stalked bacteria).
- Betaproteobacteria: mostly aerobic or facultative anaerobic bacteria that can use various organic compounds as electron donors and acceptors. Some examples are *Neisseria* (human pathogens causing gonorrhea and meningitis), *Nitrosomonas* (nitrifying bacteria), and *Burkholderia* (plant and animal pathogens).
- Gammaproteobacteria: the largest and most diverse class of proteobacteria, comprising many medically and environmentally important bacteria. Some examples are *Escherichia* (intestinal bacteria and pathogens), *Pseudomonas* (opportunistic pathogens and bioremediators), and *Vibrio* (marine bacteria and pathogens).
- Deltaproteobacteria: mostly anaerobic bacteria that are involved in sulfur and iron cycling. Some examples are *Desulfovibrio* (sulfate-reducing bacteria), *Bdellovibrio* (predatory bacteria), and *Myxococcus* (gliding bacteria that form fruiting bodies).
- Epsilonproteobacteria: mostly microaerophilic or anaerobic bacteria that inhabit extreme environments such as hot springs, hydrothermal vents, and animal guts. Some examples are *Helicobacter* (human pathogens causing ulcers and gastritis), *Campylobacter* (animal pathogens causing diarrhea), and *Wolinella* (hydrogen-oxidizing bacteria).

## Nonproteobacteria Gram-Negative Bacteria

- Nonproteobacteria are a diverse group of bacteria that do not belong to the phylum Proteobacteria, which contains the majority of gram-negative bacteria.
- Nonproteobacteria can be either gram-negative or gram-positive, depending on the presence or absence of an outer membrane in their cell wall.
- Nonproteobacteria can have various metabolic modes, such as photoautotrophy, chemoheterotrophy, or chemoautotrophy.
- Some examples of nonproteobacteria are:
  - **Chlamydia:** Obligate intracellular parasites that infect animals and humans, causing diseases such as trachoma, chlamydia, and lymphogranuloma venereum. They have a unique developmental cycle that involves two forms: elementary bodies and reticulate bodies.
  - **Spirochetes:** Motile, spiral-shaped bacteria that have a flexible cell wall and axial filaments for locomotion. They can be free-living or pathogenic, causing diseases such as syphilis, Lyme disease, and leptospirosis. They are difficult or impossible to culture in the laboratory.
  - **CFB group:** A phylum of bacteria that includes the genera Cytophaga, Fusobacterium, and Bacteroides. They share some similarities in their DNA sequences, but are phylogenetically diverse. They are mostly anaerobic and inhabit various environments, such as soil, water, and the human gut. They can degrade complex organic molecules, such as cellulose, chitin, and polysaccharides.
  - **Planctomycetes:** A phylum of bacteria that have unusual features, such as a lack of peptidoglycan in their cell wall, a compartmentalized cytoplasm, and the ability to bud. They are aquatic and can be found in freshwater, marine, and brackish habitats. They can perform various metabolic processes, such as nitrification, denitrification, and anammox.
  - **Phototrophic bacteria:** A group of bacteria that can use light as an energy source for biosynthesis. They can be either oxygenic or anoxygenic, depending on whether they produce oxygen or not during photosynthesis. They can also have different types of pigments, such as chlorophylls, bacteriochlorophylls, or carotenoids. Some examples of phototrophic bacteria are cyanobacteria, purple bacteria, green bacteria, and heliobacteria.

## Gram-positive bacteria

- Gram-positive bacteria are bacteria that have a **thick cell wall** composed of peptidoglycan and teichoic acids .
- Gram-positive bacteria can be distinguished from gram-negative bacteria

by the **Gram stain test**, which involves a chemical dye that stains the cell wall purple or blue .

- Gram-positive bacteria include many **important pathogens** such as Staphylococcus, Streptococcus, Bacillus, Clostridium, Corynebacterium, Listeria, and Mycobacterium .
- Gram-positive bacteria can cause various **infections** such as pneumonia, meningitis, anthrax, tetanus, diphtheria, tuberculosis, and leprosy .
- Gram-positive bacteria can also have **beneficial roles** such as producing antibiotics, probiotics, enzymes, and fermented foods.
- Gram-positive bacteria can undergo **bacterial transformation**, which is the uptake and expression of foreign DNA from the environment.
- Gram-positive bacteria are usually **susceptible** to antibiotics that target the cell wall, such as penicillin, vancomycin, and bacitracin . However, some gram-positive bacteria have developed **resistance** to these drugs, such as methicillin-resistant Staphylococcus aureus (MRSA) and vancomycin-resistant enterococci (VRE) .

## Lichens

- Lichens are **composite organisms** that consist of a **symbiotic association** of algae (usually green) or cyanobacteria and fungi (mostly ascomycetes and basidiomycetes).
- The dominant partner is the **fungus**, which gives the lichen the majority of its characteristics, from its **thallus shape** to its **fruiting bodies**. The alga or cyanobacterium provides **photosynthesis** and **nitrogen fixation** for the fungus.
- Lichens are found **worldwide** and occur in a variety of **environmental conditions**. They are **pioneer species**, among the first living things to grow on bare rock or areas denuded of life by a disaster.
- Lichens have different **growth forms**, such as crustose, foliose, fruticose, squamulose, and leprose. They also have different **reproduction methods**, such as vegetative propagation, soredia, isidia, and sexual spores.
- Lichens have various **roles** in the ecosystem, such as **bioindicators** of air quality, **decomposers** of organic matter, **soil stabilizers** and **erosion preventers**, and **food sources** and **habitats** for other organisms.
- Lichens also have various **applications** in human activities, such as **dyes**, **perfumes**, **medicines**, **litmus paper**, and **lichenometry**.

## Algae

- Algae are a large and diverse group of photosynthetic eukaryotic organisms that are typically aquatic and include seaweeds and many single-celled forms .
- Algae are not plants, but they share some characteristics with them, such as having chlorophyll, cell walls, and the ability to produce oxygen .
- Algae belong to the kingdom Protista, which is a polyphyletic group of



organisms that do not fit into any other kingdom.

- Algae have many types of life cycles, ranging from simple asexual reproduction to complex alternation of generations.
- Algae vary in size from microscopic to macroscopic, and they can be found in almost every habitat on Earth, from freshwater to marine, from soil to snow, from hot springs to polar regions.
- Algae are important for the biosphere, as they produce a large amount of oxygen, serve as food and habitat for many aquatic animals, and are sources of biofuels, bioplastics, pharmaceuticals, and other useful products.

## Protozoa

- Protozoa are **microscopic unicellular eukaryotes** that have a relatively complex internal structure and carry out complex metabolic activities .
- Some protozoa have structures for **propulsion or other types of movement**, such as flagella, cilia, or pseudopodia .
- Protozoa are either **free-living or parasitic**, that feed on organic matter such as other microorganisms or organic tissues and debris .
- Protozoa are **heterotrophic**, using organic carbon as a source of energy.
- Protozoa have **one or more nuclei** and may have specialized organelles, such as contractile vacuoles, food vacuoles, or plastids .
- Protozoa are **nonfilamentous**, in contrast to fungi and some algae.
- Protozoa are currently classified into **six phyla** on the basis of light and electron microscopic morphology :
  - **Sarcomastigophora**, which includes flagellates and amoebae.
  - **Apicomplexa**, which includes intracellular parasites such as Plasmodium and Toxoplasma.
  - **Ciliophora**, which includes ciliates such as Paramecium and Stentor.
  - **Microspora**, which includes spore-forming parasites such as Encephalitozoon and Nosema.
  - **Ascetosporea**, which includes parasites of marine invertebrates such as Myxozoa and Orthonectida.
  - **Myxozoa**, which includes parasites of fish and annelids such as Myxobolus and Ceratomyxa.
- Protozoa are **diverse and ecologically important** groups of organisms, playing roles in nutrient cycling, decomposition, symbiosis, and pathogenesis .
- Protozoa can reproduce by **asexual or sexual methods**, depending on the species and environmental conditions .
- Protozoa can exhibit different forms of **life cycles**, such as haploid, diploid, or alternation of generations .
- Protozoa can be identified by their **morphological, biochemical, molecular, and immunological characteristics**, as well as by their **host range, transmission mode, and clinical manifestations** .

## Helminthes

- Helminthes are **parasitic worms** that infect various hosts, such as humans, animals, and plants .
- Helminthes can be classified into two major groups: **flatworms** (platyhelminths) and **roundworms** (nematodes) .
- Flatworms have a **flat** and **leaf-like** body shape, and include **flukes** and **tapeworms** .
- Flukes are **trematodes** that have a **sucker** at one or both ends of their body, and infect the **blood, liver, lungs, or intestines** of their hosts.
- Tapeworms are **cestodes** that have a **scolex** (head) with **hooks** and **suckers**, and a **strobila** (body) composed of **proglottids** (segments) that contain **eggs**. Tapeworms infect the **intestines** of their hosts and can cause **malnutrition, abdominal pain, and cysticercosis**.
- Roundworms have a **cylindrical** and **thread-like** body shape, and include **ascarids, hookworms, pinworms, filarias, and trichinellas** .
- Ascarids are **large** and **whitish** worms that infect the **intestines** of their hosts and can cause **intestinal obstruction, malabsorption, and pneumonia**.
- Hookworms are **small** and **reddish-brown** worms that have **hook-like** mouthparts that attach to the **intestinal wall** of their hosts and suck **blood**. Hookworms can cause **anemia, diarrhea, and malnutrition**.
- Pinworms are **tiny** and **white** worms that infect the **colon and rectum** of their hosts and cause **itching and irritation** around the **anus**.
- Filarias are **thin** and **hair-like** worms that infect the **lymphatic system** or **subcutaneous tissues** of their hosts and cause **lymphedema, elephantiasis, and river blindness**.
- Trichinellas are **microscopic** and **coiled** worms that infect the **muscles** of their hosts and cause **fever, muscle pain, and inflammation**.
- Helminthes have different **life cycles** depending on their species, but they generally involve one or more **intermediate hosts** and a **definitive host**.
- Helminthes can be detected by **microscopic examination** of **stool, blood, urine, or tissue** samples, or by **immunological tests**.
- Helminthes can be treated by **anthelmintic drugs** that target their **metabolism, nervous system, or reproduction**.
- Helminthes can be prevented by **improving sanitation, wearing protective clothing, cooking food properly, and avoiding contact** with infected hosts.

## Viral Structures

- Viruses are noncellular genetic elements that use a living cell for their replication and have an extracellular state.
- Viruses are ultramicroscopic particles containing nucleic acid surrounded by protein, and in some cases, other macromolecular components such as a membranelike envelope.

- Viruses are observed in a wide range of shapes and sizes, but these shapes are constant in a particular viral family.
- The basic structure of all viruses includes nucleic acid, either DNA (Deoxyribonucleic Acid) or RNA (Ribonucleic Acid) called a genome which is enveloped by a protein coat called a capsid .
- Many viruses have a lipoprotein bilayer that encloses the capsid, called an envelope .
- The nucleic acid of a virus contains the genetic information for the synthesis of proteins and replication.
- The capsid is composed of subunits called capsomeres, which may have different shapes and arrangements depending on the type of virus.
- The envelope may have spikes or glycoproteins that help the virus attach to the host cell or evade the host immune system .
- The two main classes of viruses are RNA viruses and DNA viruses, based on the type of nucleic acid they contain.
- RNA viruses have a single-stranded or double-stranded RNA genome, and may be positive-sense (meaning their RNA can be directly translated into proteins) or negative-sense (meaning their RNA has to be converted into a complementary positive-sense RNA before translation).
- DNA viruses have a single-stranded or double-stranded DNA genome, and may be linear or circular, and may have single or multiple segments.
- Some examples of viral shapes are icosahedral (having 20 triangular faces), helical (having a cylindrical shape), complex (having a combination of icosahedral and helical structures), and spherical (having a roughly round shape) .
- Some examples of viruses with different structures are:

| Virus                | Nucleic Acid                         | Capsid      | Envelope | Shape       |
|----------------------|--------------------------------------|-------------|----------|-------------|
| Adenovirus           | Double-stranded DNA                  | Icosahedral | No       | Icosahedral |
| Influenza virus      | Single-stranded RNA (negative-sense) | Helical     | Yes      | Spherical   |
| Herpes simplex virus | Double-stranded DNA                  | Icosahedral | Yes      | Spherical   |
| Tobacco mosaic virus | Single-stranded RNA (positive-sense) | Helical     | No       | Helical     |

| Virus            | Nucleic Acid        | Capsid                            | Envelope | Shape   |
|------------------|---------------------|-----------------------------------|----------|---------|
| Bacteriophage T4 | Double-stranded DNA | Icosahedral head and helical tail | No       | Complex |

### **Viral Multiplication**

- Viral multiplication is the process of forming new viruses by using the cellular machinery of the host cells after infection.
- Viruses have different host ranges and tissue tropisms, which determine the types of cells and organisms they can infect and multiply in.
- Viruses have different life cycles, which can be classified into two types: lytic cycle and lysogenic cycle.
- In the lytic cycle, the virus attaches to the host cell, injects its genetic material, replicates its genome and proteins, assembles new virions, and lyses the host cell to release the progeny viruses.
- In the lysogenic cycle, the virus integrates its genetic material into the host cell's genome, forming a prophage, which can remain dormant for a long time or be induced to enter the lytic cycle.
- The viral life cycle can be studied using a one-step multiplication curve, which shows the phases of inoculation, eclipse, and burst.
- The inoculation phase is when the virus attaches to the host cell and enters it.
- The eclipse phase is when the viral genome and proteins are synthesized.
- The burst phase is when the new virions are assembled and released from the host cell.
- The burst size is the number of virions produced per infected cell.
- Viruses are composed of a nucleic acid genome (DNA or RNA) and a protein coat called a capsid.
- Some viruses also have a lipid envelope derived from the host cell membrane.
- Viruses are very diverse in their shapes, sizes, and structures.

### **Role of microorganisms in the production of industrial chemicals and pharmaceuticals**

- Microorganisms are used in industry to produce a variety of organic compounds including acids, growth stimulants, and enzymes.
- Microorganisms can be genetically modified or engineered to aid in large-scale production of desired products.
- Some examples of industrial chemicals and pharmaceuticals produced by microorganisms are:
  - Amino acids: Microbes make only the L-isomer of amino acids, which are essential for protein synthesis and human nutrition. Some amino

acids, such as lysine and glutamic acid, are produced by fermentation of sugars or molasses using bacteria such as *Corynebacterium* and *Brevibacterium* .

- Citric acid: Citric acid is widely used as a food additive, preservative, and flavoring agent. It is produced by the mold *Aspergillus niger*, which converts glucose or sucrose into citric acid in a submerged aerobic fermentation process .
  - Enzymes: Enzymes are biological catalysts that speed up chemical reactions. They are used in various industries such as food, detergent, textile, leather, paper, and biofuel. Some examples of enzymes produced by microorganisms are amylases, proteases, lipases, cellulases, and xylanases. These enzymes are obtained from fungi, bacteria, and actinomycetes, such as *Aspergillus*, *Bacillus*, *Streptomyces*, and *Trichoderma* .
  - Antibiotics: Antibiotics are substances that inhibit the growth or kill other microorganisms. They are used to treat bacterial infections and prevent diseases. Most antibiotics are derived from natural sources, such as fungi and bacteria. Some examples of antibiotics produced by microorganisms are penicillin, streptomycin, tetracycline, erythromycin, and cephalosporin .
  - Vitamins: Vitamins are organic compounds that are essential for normal metabolism and growth. They are obtained from food or dietary supplements. Some vitamins, such as B12, B2, and K, are produced by microorganisms, such as *Pseudomonas*, *Bacillus*, and *Propionibacterium* .
  - Biofuels: Biofuels are renewable sources of energy that are derived from biological materials, such as plants, algae, and microorganisms. Some examples of biofuels produced by microorganisms are ethanol, biodiesel, biogas, and biohydrogen. These biofuels are obtained by fermentation, transesterification, anaerobic digestion, and photosynthesis, using microorganisms such as *Saccharomyces*, *Escherichia*, *Methanobacterium*, and *Chlamydomonas* .
- In the industrial process, microorganisms act like chemical factories. To be effective, they should liberate a large amount of a single product that can be efficiently isolated and purified. The microorganisms should be easy to maintain and cultivate, and should have genetic stability with infrequent mutations.
  - The production of industrial chemicals and pharmaceuticals by microorganisms involves several steps, such as strain selection, media optimization, fermentation, downstream processing, and quality control .
  - The advantages of using microorganisms for industrial production are:
    - They are cheap and abundant sources of raw materials.
    - They have high specificity and selectivity for the desired products.
    - They can operate under mild conditions of temperature and pressure.

- They can produce complex molecules that are difficult to synthesize chemically.
- They can be genetically modified or engineered to enhance their productivity and diversity .
- The challenges of using microorganisms for industrial production are:
  - They are susceptible to contamination and infection by other microorganisms.
  - They may produce unwanted by-products or toxins that affect the quality and safety of the products.
  - They may lose their productivity or stability due to genetic mutations or environmental changes.
  - They may face ethical and social issues regarding the use of genetically modified or engineered organisms .

## **Unit 2 - Functional Anatomy of Prokaryotic and Eukaryotic Cells: Size, shape, and arrangement of bacterial cells.**

### **Structure and function of cells**

- Prokaryotic cells are cells that lack a nucleus and other membrane-bound organelles. They are usually smaller and simpler than eukaryotic cells, which have a nucleus and other organelles.
- Eukaryotic cells are cells that have a nucleus and other membrane-bound organelles. They are usually larger and more complex than prokaryotic cells, which lack a nucleus and other organelles.
- Bacterial cells are prokaryotic cells that belong to the domain Bacteria. They have a variety of shapes, sizes, and arrangements, depending on their species and environmental conditions.
- The most common shapes of bacterial cells are:
  - Coccus: spherical or oval
  - Bacillus: rod-shaped or cylindrical
  - Spirillum: spiral or curved
  - Vibrio: comma-shaped
  - Coccobacillus: short and plump
  - Spirochete: flexible and helical
- The most common arrangements of bacterial cells are:
  - Single: individual cells
  - Diplo: pairs of cells
  - Strepto: chains of cells
  - Staphylo: clusters of cells
  - Tetrads: groups of four cells
  - Sarcinae: groups of eight cells
  - Palisades: side-by-side rows of cells
- The structure and function of bacterial cells are determined by their cell envelope, which consists of three layers: the cell wall, the cell membrane,

and the capsule.

- The cell wall is a rigid layer that surrounds the cell membrane and provides shape, support, and protection to the cell. It also prevents the cell from bursting due to osmotic pressure. The cell wall is composed of peptidoglycan, a polymer of sugar and amino acids, and may also contain other substances, such as lipopolysaccharide, teichoic acid, or mycolic acid, depending on the type of bacteria. Bacteria can be classified as gram-positive or gram-negative based on the structure and staining properties of their cell wall.
- The cell membrane is a thin, flexible layer that encloses the cytoplasm and regulates the passage of materials in and out of the cell. It is composed of phospholipids and proteins, and may also contain hopanoids, which are sterol-like molecules that provide stability and fluidity to the membrane. The cell membrane is involved in many functions, such as respiration, photosynthesis, secretion, transport, and signal transduction.
- The capsule is a sticky layer that surrounds the cell wall and the cell membrane and provides protection, adhesion, and virulence to the cell. It is composed of polysaccharides, proteins, or both, and may vary in thickness and composition, depending on the type of bacteria. The capsule helps the cell to evade the immune system, attach to surfaces, and form biofilms.

### Functional Anatomy of Prokaryotic and Eukaryotic Cells

- Prokaryotic and eukaryotic cells are the two basic types of cells that make up all living organisms .
- Prokaryotic cells are simpler and smaller than eukaryotic cells, and lack a nucleus and other membrane-bound organelles .
- Eukaryotic cells are more complex and larger than prokaryotic cells, and have a nucleus that contains the DNA and other membrane-bound organelles that perform specific functions .
- The main differences between prokaryotic and eukaryotic cells are summarized in the table below:

| Feature    | Prokaryotic cells         | Eukaryotic cells                                                                           |
|------------|---------------------------|--------------------------------------------------------------------------------------------|
| Size       | 0.1-5 micrometers         | 10-100 micrometers                                                                         |
| Nucleus    | Absent                    | Present                                                                                    |
| DNA        | Circular, in cytoplasm    | Linear, in nucleus                                                                         |
| Histones   | Absent                    | Present                                                                                    |
| Organelles | Absent (except ribosomes) | Present<br>(e.g. mitochondria, chloroplasts, endoplasmic reticulum, Golgi apparatus, etc.) |

| Feature       | Prokaryotic cells                                                                 | Eukaryotic cells                                                                                                                |
|---------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Cell wall     | Present (made of peptidoglycan in bacteria, and other polysaccharides in archaea) | Absent in animals, present in plants (made of cellulose), fungi (made of chitin), and some protists (made of various materials) |
| Cell division | Binary fission                                                                    | Mitosis or meiosis                                                                                                              |
| Ribosomes     | 70S                                                                               | 80S                                                                                                                             |

- The size, shape, and arrangement of bacterial cells vary widely, and can be used to identify and classify different types of bacteria .
- Some common shapes of bacterial cells are:
  - Coccus (spherical)
  - Bacillus (rod-shaped)
  - Coccobacillus (short and plump)
  - Vibrio (curved rod)
  - Spirillum (rigid spiral)
  - Spirochete (flexible spiral)
  - Pleomorphic (variable shape)
- Some common arrangements of bacterial cells are:
  - Single (individual cells)
  - Pairs (two cells attached)
  - Chains (multiple cells attached end to end)
  - Tetrads (groups of four cells in a square)
  - Sarcinae (groups of eight cells in a cube)
  - Staphylococci (clusters of cells in a grape-like structure)
  - Palisades (cells arranged side by side in a picket fence-like structure)
- The structure and function of prokaryotic and eukaryotic cells are determined by the interactions of their components, such as the plasma membrane, cytoplasm, ribosomes, DNA, and organelles .
- The plasma membrane is a thin layer of phospholipids and proteins that surrounds the cell and regulates the movement of substances in and out of the cell .
- The cytoplasm is the fluid-filled space inside the cell that contains the DNA, ribosomes, and other molecules and structures .
- The ribosomes are the sites of protein synthesis, where the genetic information in the DNA is translated into amino acid sequences .
- The DNA is the genetic material that carries the instructions for the cell's structure and function .



- The organelles are specialized structures that perform specific functions in the cell, such as energy production, photosynthesis, protein modification, transport, and degradation .
- The cell wall is a rigid layer that surrounds the plasma membrane and provides support and protection to the cell .

### Size, shape, and arrangement of bacterial cells

- Bacterial cells are microscopic organisms that cannot be seen by the naked eye. They are measured in micrometers ( $\mu\text{m}$ ), which are one-thousandth of a millimeter. The average size of a bacterial cell varies between 0.2 and 2.0  $\mu\text{m}$  .
- Bacterial cells have different shapes, such as spherical (cocci), rod-shaped (bacilli), spiral (spirilla), comma-shaped (vibrios), or corkscrew-shaped (spirochetes). Some bacteria can change their shape depending on the environment, such as pleomorphic bacteria .
- Bacterial cells can also arrange themselves in different patterns, such as single, pairs, chains, clusters, packets, or palisades. The arrangement depends on the plane of division, the number of divisions, and the degree of separation of the cells after division .
- Some examples of bacterial shapes and arrangements are:
  - Cocci: spherical bacteria that can be single, diplococci (pairs), streptococci (chains), staphylococci (clusters), tetrads (packets of four), or sarcinae (packets of eight).
  - Bacilli: rod-shaped bacteria that can be single, diplobacilli (pairs), streptobacilli (chains), or coccobacilli (short and oval).
  - Spirilla: rigid spiral bacteria that can be single or in short chains.
  - Vibrios: comma-shaped bacteria that can be single or in pairs.
  - Spirochetes: flexible corkscrew-shaped bacteria that can be single or in clusters.
- The following table summarizes the size, shape, and arrangement of some common bacterial species :

| Bacterial species             | Size ( $\mu\text{m}$ ) | Shape      | Arrangement           |
|-------------------------------|------------------------|------------|-----------------------|
| <i>Escherichia coli</i>       | 0.5-1.0 x 2.0-6.0      | Bacillus   | Single or in pairs    |
| <i>Staphylococcus aureus</i>  | 0.5-1.0                | Coccus     | Clusters              |
| <i>Streptococcus pyogenes</i> | 0.5-1.0                | Coccus     | Chains                |
| <i>Neisseria gonorrhoeae</i>  | 0.6-1.0                | Coccus     | Diplococci            |
| <i>Bacillus anthracis</i>     | 1.0-1.2 x 3.0-10.0     | Bacillus   | Chains                |
| <i>Clostridium tetani</i>     | 0.4-0.6 x 2.0-20.0     | Bacillus   | Single or in pairs    |
| <i>Helicobacter pylori</i>    | 0.5-1.0 x 2.5-5.0      | Vibrio     | Single or in pairs    |
| <i>Treponema pallidum</i>     | 0.1-0.2 x 5.0-15.0     | Spirochete | Single or in clusters |

| Bacterial species           | Size ( $\mu\text{m}$ ) | Shape      | Arrangement               |
|-----------------------------|------------------------|------------|---------------------------|
| <i>Borrelia burgdorferi</i> | 0.2-0.3 x 5.0-20.0     | Spirochete | Single or in clusters     |
| <i>Campylobacter jejuni</i> | 0.2-0.5 x 0.5-5.0      | Spirillum  | Single or in short chains |

### Size, shape, and arrangement of bacterial cells

- Bacterial cells are microscopic organisms that cannot be seen by the naked eye. They are measured in micrometers ( $\mu\text{m}$ ), which are one-thousandth of a millimeter. The average size of a bacterial cell varies between 0.2 and 2.0  $\mu\text{m}$ .
- Bacterial cells have different shapes and arrangements that help them adapt to different environments and functions. The most common shapes are:
  - Cocci: spherical or oval cells that can be arranged in pairs (diplococci), chains (streptococci), clusters (staphylococci), tetrads (groups of four), sarcinae (cubical packets of eight), or other patterns.
  - Bacilli: rod-shaped or cylindrical cells that can be arranged in singles, pairs (diplobacilli), chains (streptobacilli), or palisades (side-by-side parallel rows).
  - Spirilla: rigid, helical or spiral-shaped cells that can be arranged in singles or pairs.
  - Spirochetes: flexible, corkscrew-shaped cells that can move by rotating their axial filaments.
  - Vibrios: comma-shaped or curved cells that can be arranged in singles or pairs.
  - Coccobacilli: oval-shaped cells that are intermediate between cocci and bacilli.
  - Filamentous: long, thin, thread-like cells that can form branching networks or mycelia.
- Some bacterial cells can change their shape and arrangement depending on the environmental conditions or the stage of their life cycle. This is called pleomorphism.
- Some bacterial cells can form specialized structures such as endospores, capsules, flagella, pili, or fimbriae that enhance their survival, attachment, or motility.

### Arrangement of Bacterial Cells

- Bacteria are microscopic, single-celled organisms that can have different shapes and arrangements.
- The shape of a bacterial cell is determined by its cell wall, which is made of peptidoglycan, a polymer of carbohydrates and proteins.
- The arrangement of a bacterial cell is determined by the plane of division during binary fission, which is the process of cell division in bacteria.
- The most common shapes of bacteria are:

- Coccus: spherical or oval
- Bacillus: rod-shaped or cylindrical
- Spiral: curved or twisted
- The most common arrangements of bacteria are:
  - Diplo: pairs of cells
  - Strepto: chains of cells
  - Staphylo: grape-like clusters of cells
  - Tetrad: groups of four cells
  - Sarcina: groups of eight cells
- Some examples of bacteria with different shapes and arrangements are:
  - Diplococci: pairs of cocci, such as *Neisseria gonorrhoeae*, the cause of gonorrhea
  - Streptococci: chains of cocci, such as *Streptococcus pyogenes*, the cause of strep throat
  - Staphylococci: clusters of cocci, such as *Staphylococcus aureus*, the cause of skin infections
  - Bacilli: single or paired rods, such as *Escherichia coli*, a common gut bacterium
  - Streptobacilli: chains of rods, such as *Bacillus anthracis*, the cause of anthrax
  - Spirilla: rigid spirals, such as *Spirillum minus*, the cause of rat-bite fever
  - Spirochetes: flexible spirals, such as *Treponema pallidum*, the cause of syphilis
- The shape and arrangement of bacteria can help in their identification and classification, as well as in understanding their function and pathogenicity.

### Structure and function of cells

- Cells are the basic units of life and the smallest living things that can perform all the functions of life.
- Cells can be classified into two major types: prokaryotic and eukaryotic.
- Prokaryotic cells are simpler and smaller than eukaryotic cells. They do not have a nucleus or other membrane-bound organelles. They have a single circular chromosome, a cell wall, a plasma membrane, ribosomes, and sometimes a capsule, flagella, or pili.
- Eukaryotic cells are more complex and larger than prokaryotic cells. They have a nucleus that contains linear chromosomes, and various membrane-bound organelles that perform specific functions. They include animals, plants, fungi, and protists.
- All cells have some common features, such as a cell membrane, cytoplasm, ribosomes, and genetic material. The cell membrane is a thin layer that separates the inside and the outside of the cell, and controls what goes in and comes out. The cytoplasm is a jelly-like substance that contains the cell's parts. The ribosomes are the sites of protein synthesis. The genetic material is the information that directs the cell's activities.

- Some of the organelles found in eukaryotic cells are:
  - Nucleus: the control center of the cell that contains the DNA and regulates gene expression.
  - Mitochondria: the powerhouses of the cell that produce energy by breaking down glucose.
  - Chloroplasts: the sites of photosynthesis in plant cells that convert light energy into chemical energy.
  - Endoplasmic reticulum: a network of membranes that transports and modifies proteins and lipids.
  - Golgi apparatus: a stack of flattened membranes that sorts and packages proteins and lipids for export.
  - Lysosomes: the garbage disposals of the cell that digest and recycle waste materials.
  - Cytoskeleton: a framework of protein filaments that provides shape, support, and movement to the cell.
  - Vacuoles: the storage compartments of the cell that hold water, nutrients, or waste products.
  - Cell wall: a rigid layer that surrounds the plasma membrane of plant cells and provides support and protection.
  - Cilia and flagella: hair-like or whip-like structures that extend from the cell surface and enable movement or sensing.

### **Unit 3 - Catabolic & anabolic reactions: enzymes, energy production and carbohydrate metabolism. Lipid & protein catabolism, Energy production mechanism, metabolic diversity & pathways of energy use. Integration of metabolism**

- Catabolic reactions are those that break down complex molecules into simpler ones, releasing energy in the process. Anabolic reactions are those that build up complex molecules from simpler ones, consuming energy in the process.
- Enzymes are biological catalysts that speed up the rate of chemical reactions by lowering the activation energy. They are highly specific for their substrates and can be regulated by various factors such as temperature, pH, inhibitors, activators, and feedback mechanisms.
- Energy production in living cells involves the transfer of electrons from reduced molecules (such as glucose, fatty acids, and amino acids) to electron acceptors (such as oxygen,  $\text{NAD}^+$ , and FAD) in a series of oxidation-reduction reactions. The energy released from these reactions is used to synthesize ATP, the universal energy currency of the cell.
- Carbohydrate metabolism is the process of converting carbohydrates (such as glucose, glycogen, starch, and cellulose) into other forms of energy or storage molecules. The main pathways of carbohydrate metabolism are glycolysis, gluconeogenesis, glycogenesis, glycogenolysis, pentose phos-

phate pathway, and citric acid cycle.

- Lipid catabolism is the process of breaking down lipids (such as triglycerides, phospholipids, and cholesterol) into smaller molecules (such as glycerol, fatty acids, and acetyl-CoA) that can be used for energy production or biosynthesis. The main pathways of lipid catabolism are lipolysis, beta-oxidation, ketogenesis, and cholesterol metabolism.
- Protein catabolism is the process of breaking down proteins (such as enzymes, hormones, and antibodies) into amino acids that can be used for energy production or biosynthesis. The main pathways of protein catabolism are proteolysis, deamination, transamination, and urea cycle.
- Energy production mechanism is the process of coupling the electron transport chain (ETC) and oxidative phosphorylation (OXPHOS) to generate ATP from the oxidation of NADH and FADH<sub>2</sub>. The ETC consists of four complexes of proteins and coenzymes that transfer electrons from NADH and FADH<sub>2</sub> to oxygen, forming water and a proton gradient across the inner mitochondrial membrane. The OXPHOS consists of a fifth complex of proteins and coenzymes that use the proton gradient to drive the synthesis of ATP from ADP and Pi.
- Metabolic diversity is the ability of different organisms to use different sources of energy and carbon for their growth and survival. Some examples of metabolic diversity are:
  - Phototrophs: use light as the source of energy
  - Chemotrophs: use chemical compounds as the source of energy
  - Autotrophs: use carbon dioxide as the source of carbon
  - Heterotrophs: use organic compounds as the source of carbon
  - Aerobes: use oxygen as the final electron acceptor
  - Anaerobes: use other molecules (such as nitrate, sulfate, or carbon dioxide) as the final electron acceptor
- Pathways of energy use are the processes of utilizing the energy stored in ATP for various cellular functions. Some examples of pathways of energy use are:
  - Biosynthesis: the synthesis of macromolecules (such as nucleic acids, proteins, lipids, and polysaccharides) from simpler precursors
  - Transport: the movement of molecules and ions across membranes (such as active transport, endocytosis, and exocytosis)
  - Mechanical work: the generation of force and movement (such as muscle contraction, flagellar rotation, and cytoskeletal dynamics)
  - Signal transduction: the transmission and amplification of signals from the extracellular environment to the intracellular response (such as hormone-receptor binding, second messenger production, and gene expression)
- Integration of metabolism is the coordination and regulation of the various metabolic pathways to maintain the homeostasis and adaptability of the cell. The integration of metabolism is achieved by various mechanisms such as:
  - Hormonal regulation: the modulation of metabolic enzymes and

- transporters by hormones (such as insulin, glucagon, and adrenaline) that are secreted by endocrine glands in response to the physiological state of the organism
- Allosteric regulation: the modulation of metabolic enzymes by small molecules (such as ATP, ADP, AMP, NAD<sup>+</sup>, NADH, acetyl-CoA, and citrate) that bind to their regulatory sites and affect their activity

### Catabolic and Anabolic Reactions

- Catabolic and anabolic reactions are two types of metabolic reactions that occur in living cells.
- Catabolic reactions are those that break down complex molecules into simpler ones, releasing energy in the process. An example of a catabolic reaction is glycolysis, which breaks down glucose into pyruvate and produces ATP.
- Anabolic reactions are those that build up complex molecules from simpler ones, using energy in the process. An example of an anabolic reaction is protein synthesis, which links amino acids together to form proteins.
- Catabolic and anabolic reactions are often coupled, meaning that the energy released by catabolic reactions is used to power anabolic reactions. A common molecule that mediates this coupling is ATP, which is produced by catabolic reactions and consumed by anabolic reactions.
- Catabolic and anabolic reactions are essential for the growth, maintenance, and repair of cells and tissues. They also regulate the balance of energy and matter in the body.
- Catabolic and anabolic reactions can be classified into different pathways based on the type of molecules involved and the location of the reactions. Some examples of catabolic pathways are glycolysis, Krebs cycle, electron transport chain, and oxidative phosphorylation, which are involved in carbohydrate metabolism and energy production. Some examples of anabolic pathways are gluconeogenesis, glycogenesis, photosynthesis, and lipogenesis, which are involved in carbohydrate synthesis and storage .
- Catabolic and anabolic reactions can also vary depending on the metabolic diversity of different organisms and tissues. For example, some bacteria can use different sources of energy and carbon, such as light, hydrogen, sulfur, or methane, to perform catabolic and anabolic reactions. Some tissues, such as muscle, liver, and brain, have different preferences and capacities for using glucose, fatty acids, or amino acids as fuel for catabolic and anabolic reactions .
- Catabolic and anabolic reactions are integrated and coordinated by various hormones, enzymes, and signaling molecules that respond to the internal and external conditions of the organism. For example, insulin and glucagon are hormones that regulate the balance of glucose and glycogen in the blood and tissues. Enzymes are proteins that catalyze and regulate the rate of catabolic and anabolic reactions. Signaling molecules are chemicals that transmit information and signals between cells and tissues

to coordinate their metabolic activities .

## **Enzymes**

- Enzymes are biological catalysts that speed up the rate of biochemical reactions by lowering the activation energy.
- Enzymes are highly specific for their substrates, which are the molecules that bind to the enzyme's active site and undergo a chemical change.
- Enzymes can be classified into two main types based on the direction of the biochemical reaction they catalyze: anabolic and catabolic enzymes.

## **Anabolic Enzymes**

- Anabolic enzymes catalyze the biochemical reactions that synthesize larger complex molecules from smaller units.
- Anabolic reactions consume energy and are also called biosynthetic reactions.
- An example of an anabolic reaction is photosynthesis, where plants make glucose molecules from different raw materials such as carbon dioxide and water.
- Anabolic enzymes are involved in the synthesis of macromolecules such as carbohydrates, lipids, proteins and nucleic acids.

## **Catabolic Enzymes**

- Catabolic enzymes catalyze the biochemical reactions that break down larger complex molecules into smaller units.
- Catabolic reactions release energy and are also called degradative reactions.
- An example of a catabolic reaction is the process of food digestion, where different enzymes break down food particles so they can be absorbed by the small intestine.
- Catabolic enzymes are involved in the breakdown of macromolecules such as carbohydrates, lipids, proteins and nucleic acids.

## **Metabolic Pathways**

- Metabolic pathways are sequences of biochemical reactions that are facilitated by enzymes.
- Metabolic pathways can be either anabolic or catabolic, depending on whether they synthesize or degrade molecules.
- Metabolic pathways can be reversible or irreversible, depending on whether the reactions can proceed in both directions or only in one direction.
- Metabolic pathways can be regulated by various factors such as the availability of substrates, the concentration of products, the presence of inhibitors or activators, and the feedback mechanisms.

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**Energy production for the notes of the Unit 3 - Catabolic & anabolic reactions: enzymes, energy production and carbohydrate metabolism. Lipid & protein catabolism, Energy production mechanism, metabolic diversity & pathways of energy use. Integration of metabolism in the subject of REMEDIAL BIOLOGY-II KCS**

- Energy production is the process of converting different sources of energy, such as chemical, light, or mechanical energy, into a form that can be used by living cells, such as adenosine triphosphate (ATP).
- Energy production involves different types of metabolic pathways, which are sequences of enzyme-catalyzed reactions that transform one or more starting molecules into products.
- Metabolic pathways can be classified into two main types: catabolic and anabolic.
  - Catabolic pathways are those that break down large molecules into smaller ones, releasing energy in the process. Examples of catabolic pathways are glycolysis, which breaks down glucose into pyruvate, and cellular respiration, which oxidizes pyruvate into carbon dioxide and water, generating ATP.
  - Anabolic pathways are those that synthesize large molecules from smaller ones, requiring energy in the process. Examples of anabolic pathways are photosynthesis, which converts light energy into chemical energy and builds glucose from carbon dioxide and water, and biosynthesis, which uses ATP and other precursors to make macromolecules such as proteins, lipids, and nucleic acids.
- Enzymes are important for catalyzing all types of metabolic pathways, as they lower the activation energy and increase the rate of the reactions. Enzymes are specific for their substrates and products, and can be regulated by different factors, such as temperature, pH, inhibitors, and activators.
- Energy production and carbohydrate metabolism are closely related, as carbohydrates are the main source of energy for most living organisms. Carbohydrates can be stored as glycogen in animals or starch in plants, and can be mobilized when needed to produce ATP through glycolysis and cellular respiration. Carbohydrates can also be converted into other molecules, such as amino acids, fatty acids, and nucleotides, through different anabolic pathways.
- Lipid and protein catabolism are alternative ways of producing energy when carbohydrates are not available or sufficient. Lipids can be broken



down into glycerol and fatty acids, which can enter glycolysis and cellular respiration, respectively. Proteins can be broken down into amino acids, which can be deaminated and converted into intermediates of glycolysis or the citric acid cycle, depending on their structure.

- Energy production mechanism is the process of coupling the energy released from catabolic pathways to the energy required for anabolic pathways, mainly through the synthesis and hydrolysis of ATP. ATP is the universal energy currency of the cell, as it can transfer its phosphate group to other molecules, making them more reactive and able to participate in different reactions. ATP is synthesized by three main mechanisms: substrate-level phosphorylation, oxidative phosphorylation, and photophosphorylation.
  - Substrate-level phosphorylation is the direct transfer of a phosphate group from a substrate to ADP, forming ATP. This occurs in some reactions of glycolysis and the citric acid cycle, as well as in the fermentation pathways that regenerate  $\text{NAD}^+$  from NADH.
  - Oxidative phosphorylation is the indirect synthesis of ATP from ADP and inorganic phosphate, using the energy of a proton gradient across a membrane. This occurs in the electron transport chain and the ATP synthase complex, which are located in the inner mitochondrial membrane in eukaryotes and the plasma membrane in prokaryotes. The electron transport chain transfers electrons from NADH and  $\text{FADH}_2$ , which are generated in glycolysis and the citric acid cycle, to oxygen, which is the final electron acceptor. The energy released from this process is used to pump protons from the matrix to the intermembrane space, creating a proton gradient. The ATP synthase complex allows the protons to flow back to the matrix, using the energy of this process to synthesize ATP from ADP and inorganic phosphate.
  - Photophosphorylation is the synthesis of ATP from ADP and inorganic phosphate, using the energy of light. This occurs in the photosystems and the ATP synthase complex, which are located in the thylakoid membrane of the chloroplasts in plants and some algae. The photosystems absorb light and transfer electrons from water to  $\text{NADP}^+$ , which is the final electron acceptor. The energy released from

## Carbohydrate Metabolism

- Carbohydrate metabolism is the whole of the biochemical processes responsible for the metabolic formation, breakdown, and interconversion of carbohydrates in living organisms.
- Carbohydrates are organic molecules composed of carbon, hydrogen, and oxygen atoms. They can be classified into monosaccharides (simple sugars), disaccharides (two monosaccharides linked by a glycosidic bond), oligosaccharides (three to ten monosaccharides), and polysaccharides

(more than ten monosaccharides).

- The most common and important monosaccharide in living organisms is glucose, which is the main source of energy for cells. Other monosaccharides include fructose, galactose, ribose, and deoxyribose.
- The most common and important disaccharides in living organisms are sucrose (glucose + fructose), lactose (glucose + galactose), and maltose (glucose + glucose).
- The most common and important polysaccharides in living organisms are starch (a storage form of glucose in plants), glycogen (a storage form of glucose in animals), and cellulose (a structural component of plant cell walls).
- Carbohydrate metabolism begins with digestion in the small intestine where monosaccharides are absorbed into the blood stream. Blood sugar concentrations are controlled by three hormones: insulin, glucagon, and epinephrine.
- Insulin is secreted by the pancreas when the blood glucose level is high, and it stimulates the uptake of glucose by cells, especially in the liver, muscle, and adipose tissue. Insulin also promotes the synthesis of glycogen, fat, and protein, and inhibits the breakdown of these molecules.
- Glucagon is secreted by the pancreas when the blood glucose level is low, and it stimulates the breakdown of glycogen in the liver and the release of glucose into the blood. Glucagon also promotes the breakdown of fat and protein, and inhibits the synthesis of these molecules.
- Epinephrine is secreted by the adrenal glands in response to stress, and it enhances the effects of glucagon and inhibits the effects of insulin. Epinephrine also stimulates the breakdown of glycogen in the muscle and the release of glucose and fatty acids into the blood.
- The main pathway of glucose metabolism in cells is glycolysis, which is the breakdown of glucose into two molecules of pyruvate, with the net production of two molecules of ATP (adenosine triphosphate) and two molecules of NADH (nicotinamide adenine dinucleotide) per glucose molecule.
- Pyruvate can have different fates depending on the availability of oxygen and the type of cell. In the presence of oxygen, pyruvate enters the mitochondria and is converted into acetyl-CoA (acetyl coenzyme A), which then enters the citric acid cycle (also known as the Krebs cycle or the tricarboxylic acid cycle). The citric acid cycle is a series of reactions that produces two molecules of CO<sub>2</sub> (carbon dioxide), one molecule of ATP, three molecules of NADH, and one molecule of FADH<sub>2</sub> (flavin adenine dinucleotide) per acetyl-CoA molecule.
- The NADH and FADH<sub>2</sub> molecules generated by glycolysis and the citric acid cycle are then used to produce more ATP by a process called oxidative phosphorylation, which involves the transfer of electrons from these molecules to oxygen through a series of protein complexes called the electron transport chain. The electron transport chain creates a proton gradient across the inner mitochondrial membrane, which drives the synthesis of ATP by a protein called ATP synthase.

- In the absence of oxygen, or in some cells that lack mitochondria, pyruvate is converted into lactate (lactic acid) by a process called anaerobic glycolysis, which regenerates NAD<sup>+</sup> (nicotinamide adenine dinucleotide) from NADH and allows glycolysis to continue. Lactate can be transported to the liver and converted back into glucose by a process called gluconeogenesis, which is the synthesis of glucose from non-carbohydrate sources.
- Gluconeogenesis can also occur in the liver and the kidney from other substrates, such as amino acids, glycerol, and lactate, when the blood glucose level is low or during fasting,

### **Lipid and Protein Catabolism**

- Lipids and proteins are important macromolecules that can be used as energy sources by heterotrophic organisms.
- Lipids are composed of glycerol and fatty acids, while proteins are composed of amino acids.
- Lipids and proteins are broken down by extracellular enzymes (lipases and proteases) into smaller molecules that can be transported into the cell.
- Lipid catabolism involves the following steps :
  - Lipases hydrolyze lipids into glycerol and free fatty acids.
  - Glycerol is phosphorylated and converted into dihydroxyacetone phosphate (DHAP), which can enter glycolysis or gluconeogenesis.
  - Fatty acids are activated by coenzyme A (CoA) and transported into the mitochondria by carnitine shuttle.
  - Fatty acids undergo beta-oxidation, which produces acetyl-CoA, NADH, and FADH<sub>2</sub>.
  - Acetyl-CoA can enter the citric acid cycle or ketogenesis, while NADH and FADH<sub>2</sub> can be used for oxidative phosphorylation.
- Protein catabolism involves the following steps :
  - Proteases cleave proteins into smaller peptides and amino acids.
  - Peptides and amino acids are transported into the cell by specific transporters.
  - Amino acids undergo deamination, which removes the amino group and produces ammonia and keto acids.
  - Ammonia is excreted or converted into urea or uric acid, depending on the organism.
  - Keto acids can be converted into pyruvate, acetyl-CoA, or intermediates of the citric acid cycle, depending on the type of amino acid.
  - Pyruvate, acetyl-CoA, and citric acid cycle intermediates can be used for energy production or biosynthesis.

### **Energy production mechanism**

- Energy production is the process of converting energy from one form to another in living systems.
- Energy comes to living systems through electrons occupying high energy

states, either from food (respiratory chains) or from light (photosynthesis).

- Energy is exchanged between living systems and their surroundings as they use energy from the sun to perform photosynthesis or consume energy-storing molecules and release energy to the environment by doing work and releasing heat.
- Energy is subject to physical laws, such as the first law of thermodynamics, which states that energy cannot be created or destroyed, only transformed, and the second law of thermodynamics, which states that the entropy (disorder) of the universe tends to increase.
- The primary energy-releasing pathways of metabolism involve the breakdown and synthesis of carbohydrates, lipids, and amino acids.
- Carbohydrates are the most important energy source for most living organisms. They are broken down into glucose, which is then oxidized through glycolysis, the citric acid cycle, and oxidative phosphorylation to produce ATP, the universal energy currency of the cell .
- Lipids are also an important energy source, especially for animals that store fat as an energy reserve. They are broken down into glycerol and fatty acids, which can enter glycolysis or the citric acid cycle, respectively, to generate ATP.
- Amino acids are the building blocks of proteins, but they can also be used as an energy source when carbohydrates and lipids are scarce. They are broken down into various intermediates that can enter glycolysis or the citric acid cycle, depending on the type of amino acid.
- Energy production can also vary depending on the availability of oxygen. Aerobic respiration is the process of using oxygen as the final electron acceptor in the electron transport chain, which produces the most ATP per glucose molecule. Anaerobic respiration is the process of using other molecules, such as nitrate or sulfate, as the final electron acceptor, which produces less ATP per glucose molecule. Fermentation is the process of using organic molecules, such as pyruvate or ethanol, as the final electron acceptor, which produces the least ATP per glucose molecule.
- Energy production can also vary depending on the type of organism and the environment. Some organisms, such as plants, algae, and some bacteria, can perform photosynthesis, which is the process of using light energy to convert carbon dioxide and water into organic molecules, such as glucose, and oxygen. Some organisms, such as animals, fungi, and some bacteria, can perform cellular respiration, which is the process of using organic molecules, such as glucose, and oxygen to produce carbon dioxide, water, and ATP. Some organisms, such as archaea and some bacteria, can perform chemosynthesis, which is the process of using chemical energy, such as hydrogen sulfide or ammonia, to produce organic molecules, such as glucose, and other compounds, such as sulfur or nitrate .
- Energy production is integrated with other metabolic pathways in the cell, such as biosynthesis, degradation, and regulation. The intermediates of energy production can be used as precursors for the synthesis of other molecules, such as nucleic acids, lipids, and hormones. The products of

energy production can be used as signals for the regulation of gene expression, enzyme activity, and cellular growth. The balance between energy production and consumption is essential for the maintenance of cellular homeostasis.

### **Metabolic Diversity and Pathways of Energy Use**

- Metabolic diversity mainly refers to the different metabolic strategies that organisms have evolved to obtain energy.
- Metabolic pathways are series of connected chemical reactions that feed one another and convert starting molecules into products.
- Metabolic pathways can be broadly divided into two categories based on their effects: catabolic pathways and anabolic pathways.
- Catabolic pathways are those that break down complex molecules into simpler ones and release energy, such as glycolysis, lipid catabolism and protein catabolism .
- Anabolic pathways are those that synthesize complex molecules from simpler ones and consume energy, such as photosynthesis, gluconeogenesis and protein synthesis .
- The energy released or consumed by metabolic pathways is often stored or transferred by a molecule called adenosine triphosphate (ATP), which is the universal energy currency of cells .
- The energy production mechanism of cells depends on the availability of oxygen and the type of metabolic pathway. There are two main types of energy production mechanisms: aerobic respiration and anaerobic respiration .
- Aerobic respiration is the process of breaking down glucose (or other organic molecules) in the presence of oxygen to produce carbon dioxide, water and ATP. It involves three major stages: glycolysis, the citric acid cycle and the electron transport chain .
- Anaerobic respiration is the process of breaking down glucose (or other organic molecules) in the absence of oxygen to produce ATP and other products, such as lactic acid or ethanol. It involves only glycolysis and a fermentation step .
- Different organisms have different metabolic pathways and energy production mechanisms depending on their environment, evolutionary history and nutritional needs. For example, plants use photosynthesis to capture sunlight and convert it into chemical energy, while animals use cellular respiration to extract energy from food. Some bacteria can use inorganic molecules, such as hydrogen sulfide or nitrate, as electron donors or acceptors for energy production, while others can switch between aerobic and anaerobic respiration depending on oxygen availability. These are some examples of the metabolic diversity and pathways of energy use among living things.

### **Integration of metabolism**

- Metabolism is the sum of all the chemical reactions that take place in a living cell to maintain its structure and function.
- Metabolism can be divided into two major types: catabolism and anabolism.
- Catabolism is the breakdown of complex molecules into simpler ones, releasing energy in the process. Examples of catabolic reactions are glycolysis, lipid oxidation, and protein degradation.
- Anabolism is the synthesis of complex molecules from simpler ones, requiring energy in the process. Examples of anabolic reactions are gluconeogenesis, lipid synthesis, and protein synthesis.
- Metabolism is a continuous and dynamic process, with thousands of reactions occurring simultaneously in different parts of the cell and in different tissues of the body.
- Metabolism is regulated by various factors, such as enzyme activity, substrate availability, allosteric modulation, hormonal signals, and feedback inhibition.
- Metabolism is integrated by the coordination of different pathways and organs to achieve a balance between energy production and consumption, and between biosynthesis and degradation of biomolecules.
- Metabolism is influenced by the physiological state of the organism, such as fasting, feeding, exercise, stress, and disease.
- Metabolism is also affected by the environmental factors, such as temperature, light, oxygen, and nutrient availability.
- Metabolism is essential for the survival and adaptation of living organisms to changing conditions.

### **Unit 4 - Energy Utilization: Structure of mitochondria, cellular respiration, relationship of carbohydrate metabolism to other compounds, Glycolysis, formation of acetyl co-A, Krebs cycle, Electron Transport System and Oxidative Phosphorylation, ATP, factors affecting respiration**

- Mitochondria are organelles that are responsible for the production of most of the energy in eukaryotic cells. They have a double membrane structure: an outer membrane that is smooth and an inner membrane that is folded into cristae. The space between the two membranes is called the intermembrane space, and the space inside the inner membrane is called the matrix. The matrix contains enzymes, DNA, and ribosomes.
- Cellular respiration is the process of breaking down organic molecules, such as glucose, to release energy in the form of ATP. It involves three main stages: glycolysis, the Krebs cycle, and the electron transport system and oxidative phosphorylation. Cellular respiration can be aerobic (using oxygen) or anaerobic (without oxygen).

- Carbohydrate metabolism is the conversion of carbohydrates into other compounds, such as pyruvate, acetyl co-A, and carbon dioxide. Carbohydrate metabolism is linked to other metabolic pathways, such as lipid metabolism, protein metabolism, and nucleic acid metabolism. For example, acetyl co-A can be used to synthesize fatty acids, cholesterol, and ketone bodies, or to enter the Krebs cycle. Pyruvate can be converted into amino acids, such as alanine and glutamate, or into lactate under anaerobic conditions. Carbon dioxide can be used for photosynthesis or for the synthesis of nucleotides.
- Glycolysis is the first stage of cellular respiration, in which glucose is split into two molecules of pyruvate. Glycolysis occurs in the cytoplasm and does not require oxygen. It involves 10 enzymatic reactions, which can be divided into two phases: the preparatory phase and the payoff phase. The preparatory phase consumes 2 ATP molecules to phosphorylate glucose and fructose-6-phosphate. The payoff phase produces 4 ATP molecules and 2 NADH molecules by oxidizing glyceraldehyde-3-phosphate and reducing NAD<sup>+</sup>. The net gain of glycolysis is 2 ATP and 2 NADH per glucose molecule.
- Formation of acetyl co-A is the link between glycolysis and the Krebs cycle. It occurs in the mitochondrial matrix and requires oxygen. It involves the oxidation of pyruvate by the pyruvate dehydrogenase complex, which consists of three enzymes and five coenzymes. The products of this reaction are acetyl co-A, carbon dioxide, and NADH. Acetyl co-A can then enter the Krebs cycle or be used for other metabolic pathways.
- The Krebs cycle, also known as the citric acid cycle or the tricarboxylic acid cycle, is the second stage of cellular respiration, in which acetyl co-A is oxidized to carbon dioxide and reduced coenzymes. It occurs in the mitochondrial matrix and requires oxygen. It involves eight enzymatic reactions, which can be divided into two phases: the condensation phase and the oxidation phase. The condensation phase involves the formation of citrate from acetyl co-A and oxaloacetate. The oxidation phase involves the oxidation of citrate to oxaloacetate, producing 2 carbon dioxide, 3 NADH, 1 FADH<sub>2</sub>, and 1 ATP (or GTP) per acetyl co-A molecule. The oxaloacetate can then combine with another acetyl co-A to continue the cycle.
- The electron transport system and oxidative phosphorylation is the third and final stage of cellular respiration, in which the reduced coenzymes (NADH and FADH<sub>2</sub>) are oxidized and the energy released is used to generate ATP. It occurs in the inner mitochondrial membrane and requires oxygen. It involves four protein complexes (I, II, III, and IV) and two mobile electron carriers (ubiquinone and cytochrome c). The electrons from NADH and FADH<sub>2</sub> are transferred to the protein complexes, which pump protons from the matrix to the intermembrane space, creating a proton gradient. The protons then flow back to the matrix through the ATP synthase, which uses the proton motive force to synthesize ATP from ADP and Pi. The final electron acceptor is oxygen, which combines with

protons and electrons to form water. The theoretical yield of this stage is 32 or 34 ATP per glucose molecule, depending on the shuttle system used to transport NADH from the cytoplasm to the

### **Energy Utilization**

- Energy utilization is the process of converting the chemical energy stored in organic molecules into a usable form of energy, such as ATP, for cellular activities.
- The main pathway of energy utilization in most cells is cellular respiration, which involves the oxidation of glucose and other organic molecules to produce carbon dioxide and water, and the reduction of oxygen to form water.
- Cellular respiration consists of four stages: glycolysis, formation of acetyl co-A, Krebs cycle, and electron transport system and oxidative phosphorylation.
- Cellular respiration takes place in the cytoplasm and mitochondria of each cell of the body.

### **Structure of mitochondria**

- Mitochondria are membrane-bound organelles that are found in most eukaryotic cells, except for mature red blood cells.
- Mitochondria have two distinct membranes: an outer membrane and an inner membrane. The outer membrane is smooth and permeable to small molecules, while the inner membrane is folded into numerous projections called cristae, which increase the surface area for biochemical reactions.
- The space between the two membranes is called the intermembrane space, and the space enclosed by the inner membrane is called the matrix. The matrix contains enzymes, ribosomes, DNA, and various molecules involved in the Krebs cycle and oxidative phosphorylation.

### **Glycolysis**

- Glycolysis is the first stage of cellular respiration, and it occurs in the cytoplasm of the cell.
- Glycolysis is the breakdown of glucose, a six-carbon molecule, into two molecules of pyruvate, a three-carbon molecule, with the net production of two ATP and two NADH molecules.
- Glycolysis consists of two phases: an energy investment phase, where two ATP molecules are used to activate glucose, and an energy payoff phase, where four ATP and two NADH molecules are generated.
- Glycolysis does not require oxygen, and it can occur in both aerobic and anaerobic conditions.

### **Formation of acetyl co-A**



- Formation of acetyl co-A is the second stage of cellular respiration, and it occurs in the matrix of the mitochondria.
- Formation of acetyl co-A is the conversion of pyruvate into acetyl coenzyme A (acetyl co-A), a two-carbon molecule, with the release of carbon dioxide and the production of one NADH molecule per pyruvate.
- Formation of acetyl co-A is catalyzed by a complex of enzymes called the pyruvate dehydrogenase complex, which requires several coenzymes, such as thiamine, lipoic acid, and coenzyme A.
- Formation of acetyl co-A is an irreversible reaction, and it links glycolysis to the Krebs cycle.

### **Krebs cycle**

- Krebs cycle, also known as the citric acid cycle or the tricarboxylic acid cycle, is the third stage of cellular respiration, and it occurs in the matrix of the mitochondria.
- Krebs cycle is a series of reactions that oxidize acetyl co-A into carbon dioxide and water, and generate one ATP, three NADH, and one FADH<sub>2</sub> molecule per acetyl co-A.
- Krebs cycle consists of eight steps, each catalyzed by a specific enzyme. The first step is the condensation of acetyl co-A with oxaloacetate, a four-carbon molecule, to form citrate, a six-carbon molecule. The next seven steps involve the rearrangement and decarboxylation of citrate, until oxaloacetate is regenerated.
- Krebs cycle is a cyclic process, and it operates twice per glucose molecule, since each glucose molecule produces two acetyl co-A molecules.

### **Electron transport system and oxidative phosphorylation**

- Electron transport system and oxidative phosphorylation are the fourth and final stage of cellular respiration, and they occur in the inner membrane of the mitochondria.
- Electron transport system is a series of protein complexes and electron carriers that transfer the electrons from NADH and FADH<sub>2</sub> to oxygen, the final electron acceptor. The electron transport system creates a proton gradient across the inner membrane, by pumping protons from the matrix to the intermembrane space.
- Oxidative phosphorylation is the process of using the proton gradient to drive the synthesis of ATP by a protein complex called ATP synthase, which allows the protons to flow back to the matrix. Oxidative phosphorylation produces about 28 ATP molecules per glucose molecule, depending on the

### **Structure of mitochondria**

- Mitochondria are small, rod-shaped organelles found in both plant and animal cells that are involved in cellular respiration and energy production

- Mitochondria have a double membrane structure, consisting of an outer membrane and an inner membrane . The outer membrane is smooth and surrounds the organelle, while the inner membrane is folded into projections called cristae .
- The space between the outer and inner membranes is called the intermembrane space, and the space enclosed by the inner membrane is called the matrix . The matrix contains mitochondrial DNA, ribosomes, enzymes, and other molecules .
- The cristae increase the surface area of the inner membrane, which is where the electron transport chain and oxidative phosphorylation take place . These are the final steps of cellular respiration, which produce most of the ATP from the breakdown of glucose and other organic molecules .
- Mitochondria can vary in number, shape, and size depending on the cell type and energy demand . Cells that require a lot of energy, such as muscle cells, can contain hundreds or thousands of mitochondria. Mitochondria can also fuse and divide to adapt to changing conditions.
- Mitochondria are believed to have evolved from ancient bacteria that were engulfed by ancestral eukaryotic cells in a process called endosymbiosis . This explains why mitochondria have their own DNA, ribosomes, and membranes that are similar to those of bacteria . Mitochondrial DNA is inherited maternally in most organisms.
- Mitochondria play important roles in various cellular processes, such as apoptosis (programmed cell death), calcium signaling, heat production, and hormone synthesis. Mitochondrial dysfunction can lead to various diseases, such as diabetes, cancer, neurodegenerative disorders, and aging

## Cellular Respiration

- Cellular respiration is the process by which biological fuels are oxidised in the presence of an inorganic electron acceptor such as oxygen to produce large amounts of energy, to drive the bulk production of adenosine triphosphate (ATP).
- ATP is the main energy currency of the cell, and it is used to power various cellular processes such as biosynthesis, transport, and movement.
- Cellular respiration can be divided into four stages: glycolysis, pyruvate oxidation, the citric acid or Krebs cycle, and oxidative phosphorylation.

## Structure of mitochondria

- Mitochondria are the organelles where most of the cellular respiration takes place in eukaryotic cells.
- Mitochondria have a double membrane structure: an outer membrane that surrounds the organelle, and an inner membrane that folds into cristae.
- The space between the two membranes is called the intermembrane space,

and the space enclosed by the inner membrane is called the matrix.

- The inner membrane contains many proteins that are involved in the electron transport chain and oxidative phosphorylation.
- The matrix contains enzymes that catalyse the reactions of pyruvate oxidation and the Krebs cycle.

### **Relationship of carbohydrate metabolism to other compounds**

- Carbohydrate metabolism is the breakdown of glucose and other sugars to generate energy in the form of ATP.
- Carbohydrate metabolism is linked to other metabolic pathways that involve the synthesis or degradation of other compounds such as amino acids, lipids, and nucleotides.
- For example, some intermediates of glycolysis and the Krebs cycle can be used to synthesise amino acids, which are the building blocks of proteins.
- Similarly, some amino acids can be converted to pyruvate or acetyl-CoA, which are the substrates for the Krebs cycle.
- Lipids can also be metabolised to generate acetyl-CoA, which can enter the Krebs cycle or be used to synthesise fatty acids.
- Nucleotides can be synthesised from amino acids, sugars, and other precursors, or degraded to produce energy or nitrogenous waste.

### **Glycolysis**

- Glycolysis is the first stage of cellular respiration, and it takes place in the cytoplasm of the cell.
- Glycolysis is the process of splitting one molecule of glucose (a six-carbon sugar) into two molecules of pyruvate (a three-carbon compound).
- Glycolysis involves ten enzymatic reactions, and it can be divided into two phases: the energy investment phase and the energy payoff phase.
- In the energy investment phase, two molecules of ATP are used to phosphorylate glucose and its derivatives, making them more reactive.
- In the energy payoff phase, four molecules of ATP and two molecules of NADH (a reduced form of nicotinamide adenine dinucleotide, an electron carrier) are produced for each molecule of glucose.
- The net result of glycolysis is the production of two molecules of pyruvate, two molecules of ATP, and two molecules of NADH per molecule of glucose.

### **Formation of acetyl co-A**

- Acetyl co-A (or acetyl coenzyme A) is a two-carbon compound that is derived from pyruvate and serves as the entry point for the Krebs cycle.
- The formation of acetyl co-A from pyruvate is also known as pyruvate oxidation, and it takes place in the mitochondrial matrix.
- Pyruvate oxidation involves three steps: the decarboxylation of pyruvate, the oxidation of the remaining two-carbon fragment, and the transfer of

the acetyl group to coenzyme A.

- In the decarboxylation step, a carboxyl group is removed from pyruvate, releasing a molecule of carbon dioxide as a waste product.
- In the oxidation step, the two-carbon fragment is oxidised by a complex of enzymes called the pyruvate dehydrogenase complex, which transfers two electrons and one proton to  $\text{NAD}^+$ , forming NADH.
- In the transfer step, the acetyl group is attached to coenzyme A, a molecule that carries acyl

### Relationship of carbohydrate metabolism to other compounds

- Carbohydrate metabolism is the process of breaking down and synthesizing carbohydrates, which are organic molecules composed of carbon, hydrogen, and oxygen atoms.
- Carbohydrates can be classified into simple sugars (monosaccharides and disaccharides) and complex sugars (polysaccharides), such as starch, glycogen, and cellulose.
- Carbohydrate metabolism is connected to the metabolism of other compounds, such as proteins and lipids, through common intermediates and pathways .
- The main pathway of carbohydrate metabolism is glycolysis, which converts glucose into pyruvate, generating ATP and NADH.
- Pyruvate can then enter the mitochondria and be converted into acetyl coenzyme A (acetyl CoA), which is the entry point for the citric acid cycle (also known as the Krebs cycle or the tricarboxylic acid cycle).
- The citric acid cycle is a series of reactions that produces more ATP, NADH,  $\text{FADH}_2$ , and  $\text{CO}_2$ .
- The NADH and  $\text{FADH}_2$  then donate their electrons to the electron transport chain, which is a series of protein complexes embedded in the inner mitochondrial membrane.
- The electron transport chain transfers the electrons along a gradient of proton pumps, creating a proton motive force that drives the synthesis of ATP by the enzyme ATP synthase.
- This process is called oxidative phosphorylation, because it involves the oxidation of NADH and  $\text{FADH}_2$  and the phosphorylation of ADP to ATP.
- Carbohydrate metabolism is linked to the metabolism of other compounds in several ways:
  - Some simple sugars, such as galactose and fructose, can be converted into glucose or intermediates of glycolysis, and thus enter the same pathway .
  - Some complex sugars, such as glycogen and starch, can be broken down into glucose or glucose-1-phosphate, and thus enter the same pathway .
  - Some pentose sugars, such as ribose and deoxyribose, can be synthesized or degraded by the pentose phosphate pathway, which is connected to glycolysis by the interconversion of glucose-6-phosphate

- and ribulose-5-phosphate .
- Some amino acids, which are the building blocks of proteins, can be converted into pyruvate, acetyl CoA, or intermediates of the citric acid cycle, and thus enter the same pathway .
- Some fatty acids, which are the components of lipids, can be broken down by beta-oxidation into acetyl CoA, and thus enter the same pathway .
- Some acetyl CoA, which is the common intermediate of carbohydrate, protein, and lipid metabolism, can be used to synthesize fatty acids, cholesterol, ketone bodies, or other compounds .
- The relationship of carbohydrate metabolism to other compounds is regulated by various factors, such as the availability of substrates, the demand for energy, the hormonal signals, and the feedback inhibition .
- The relationship of carbohydrate metabolism to other compounds is important for maintaining the homeostasis of energy, glucose, and other metabolites in the body .

## Glycolysis

- Glycolysis is a metabolic pathway that converts glucose, a six-carbon sugar, into two molecules of pyruvate, a three-carbon compound.
- Glycolysis does not require oxygen and can occur in both aerobic and anaerobic conditions. In anaerobic conditions, pyruvate is further converted into lactic acid or ethanol.
- Glycolysis is an ancient metabolic pathway that is found in most living organisms. It is the first stage of cellular respiration, which produces energy in the form of ATP and NADH.
- Glycolysis consists of ten enzyme-catalyzed reactions that can be divided into two phases: the preparatory phase and the payoff phase.
- In the preparatory phase, two molecules of ATP are used to phosphorylate glucose and split it into two molecules of glyceraldehyde-3-phosphate (G3P).
- In the payoff phase, each G3P molecule is oxidized and phosphorylated, producing two molecules of NADH and four molecules of ATP. The final product is two molecules of pyruvate.
- The net gain of glycolysis is two molecules of ATP and two molecules of NADH per molecule of glucose.
- The main steps of glycolysis are summarized below:

| Step | Enzyme     | Reaction                                         | Products            |
|------|------------|--------------------------------------------------|---------------------|
| 1    | Hexokinase | Glucose + ATP<br>-> Glucose-6-phosphate +<br>ADP | Glucose-6-phosphate |

| Step | Enzyme                                   | Reaction                                                                                                         | Products                                               |
|------|------------------------------------------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| 2    | Phosphoglucose isomerase                 | Glucose-6-phosphate $\rightarrow$ Fructose-6-phosphate                                                           | Fructose-6-phosphate                                   |
| 3    | Phosphofructokinase                      | Fructose-6-phosphate + ATP $\rightarrow$ Fructose-1,6-bisphosphate + ADP                                         | Fructose-1,6-bisphosphate                              |
| 4    | Aldolase                                 | Fructose-1,6-bisphosphate $\rightarrow$ Dihydroxyacetone phosphate + Glyceraldehyde-3-phosphate                  | Dihydroxyacetone phosphate, Glyceraldehyde-3-phosphate |
| 5    | Triose phosphate isomerase               | Dihydroxyacetone phosphate $\rightarrow$ Glyceraldehyde-3-phosphate                                              | Glyceraldehyde-3-phosphate                             |
| 6    | Glyceraldehyde-3-phosphate dehydrogenase | Glyceraldehyde-3-phosphate + NAD <sup>+</sup> + Pi $\rightarrow$ 1,3-Bisphosphoglycerate + NADH + H <sup>+</sup> | 1,3-Bisphosphoglycerate, NADH                          |
| 7    | Phosphoglycerate kinase                  | 1,3-Bisphosphoglycerate + ADP $\rightarrow$ 3-Phosphoglycerate + ATP                                             | 3-Phosphoglycerate, ATP                                |
| 8    | Phosphoglycerate mutase                  | 3-Phosphoglycerate $\rightarrow$ 2-Phosphoglycerate                                                              | 2-Phosphoglycerate                                     |
| 9    | Enolase                                  | 2-Phosphoglycerate $\rightarrow$ Phosphoenolpyruvate + H <sub>2</sub> O                                          | Phosphoenolpyruvate                                    |
| 10   | Pyruvate kinase                          | Phosphoenolpyruvate + ADP $\rightarrow$ Pyruvate + ATP                                                           | Pyruvate, ATP                                          |

- The fate of pyruvate depends on the availability of oxygen. In aerobic

conditions, pyruvate is transported into the mitochondria, where it is converted into acetyl-CoA by the pyruvate dehydrogenase complex. Acetyl-CoA then enters the Krebs cycle, where it is further oxidized and produces more ATP, NADH, and FADH<sub>2</sub>. In anaerobic conditions, pyruvate is reduced into lactic acid by lactate dehydrogenase in animals, or into ethanol and carbon dioxide by alcohol dehydrogenase and pyruvate decarboxylase in plants and yeast. These reactions regenerate NAD<sup>+</sup>, which is essential for glycolysis to continue.

### Formation of Acetyl Co-A

- Acetyl Co-A (acetyl coenzyme A) is a molecule that participates in many biochemical reactions in protein, carbohydrate and lipid metabolism.
- Its main function is to deliver the acetyl group to the citric acid cycle (Krebs cycle) to be oxidized for energy production.
- Acetyl Co-A formation occurs inside or outside the cell mitochondria.
- It can be produced via the catabolism (breakdown) of carbohydrates (glucose) and lipids (fatty acids).
- Acetyl Co-A formation most commonly occurs during glucose catabolism.
- After carbohydrates have been broken down by digestive enzymes, the first stage of cellular glucose metabolism or glycolysis can begin.
- Glycolysis is the breaking down of glucose molecules into two molecules of pyruvate, a three-carbon compound, in the cytoplasm.
- Pyruvate then enters the mitochondria, where it is converted into acetyl Co-A by a complex enzyme called pyruvate dehydrogenase.
- The overall formation reaction of acetyl Co-A may be represented as: pyruvic acid + CoA + NAD<sup>+</sup> → acetyl Co-A + NADH + H<sup>+</sup> + CO<sub>2</sub>.
- This reaction may be called the oxidative decarboxylation of pyruvic acid to acetyl Co-A, because it involves the removal of a carboxyl group (CO<sub>2</sub>) and the transfer of electrons to NAD<sup>+</sup>.
- The acetyl Co-A then combines with a four-carbon molecule called oxaloacetate and goes through a cycle of reactions, ultimately regenerating the oxaloacetate.
- This cycle is called the citric acid cycle or the Krebs cycle, and it takes place in the mitochondrial matrix.
- The citric acid cycle produces ATP (or, in some cases, GTP), NADH, and FADH<sub>2</sub>, and releases carbon dioxide as a waste product.
- The NADH and FADH<sub>2</sub> then transfer their electrons to the electron transport chain, which is located in the inner mitochondrial membrane.
- The electron transport chain uses the energy from the electrons to pump protons across the membrane, creating a proton gradient.
- The protons then flow back through a protein called ATP synthase, which uses the energy to synthesize ATP from ADP and Pi.
- This process is called oxidative phosphorylation, because it involves the use of oxygen as the final electron acceptor and the production of ATP.
- The oxygen then combines with the electrons and protons to form water,

which is also released as a waste product.

- The ATP produced by the citric acid cycle, the electron transport chain, and oxidative phosphorylation is the main source of energy for cellular activities.
- The factors that affect cellular respiration include the availability of substrates (glucose, fatty acids, oxygen), the demand for ATP, the regulation of enzymes, and the environmental conditions (temperature, pH).

### Kreb cycle

- The Kreb cycle, also known as the citric acid cycle or the tricarboxylic acid cycle, is a series of chemical reactions that occur in the mitochondria of aerobic organisms .
- The Kreb cycle is the second stage of cellular respiration, following glycolysis and preceding the electron transport chain and oxidative phosphorylation .
- The Kreb cycle produces energy by oxidizing acetyl-CoA, which is derived from the breakdown of carbohydrates, fats, and proteins .
- The Kreb cycle consists of eight steps, each catalyzed by a specific enzyme :
  - Step 1: Acetyl-CoA combines with oxaloacetate to form citrate, releasing coenzyme A.
  - Step 2: Citrate is converted to isocitrate by an isomerase enzyme.
  - Step 3: Isocitrate is oxidized to alpha-ketoglutarate, releasing a molecule of carbon dioxide and a molecule of NADH.
  - Step 4: Alpha-ketoglutarate is oxidized to succinyl-CoA, releasing another molecule of carbon dioxide and NADH.
  - Step 5: Succinyl-CoA is converted to succinate, releasing a molecule of GTP (or ATP in some organisms).
  - Step 6: Succinate is oxidized to fumarate, reducing a molecule of FAD to FADH<sub>2</sub>.
  - Step 7: Fumarate is hydrated to malate by an enzyme called fumarase.
  - Step 8: Malate is oxidized to oxaloacetate, regenerating NADH and completing the cycle.
- The Kreb cycle produces two molecules of carbon dioxide, three molecules of NADH, one molecule of FADH<sub>2</sub>, and one molecule of GTP (or ATP) per acetyl-CoA molecule that enters the cycle .
- The Kreb cycle also provides intermediates for the synthesis of certain amino acids, such as aspartate, glutamate, and alanine .
- The Kreb cycle is regulated by the availability of substrates (such as acetyl-CoA and oxaloacetate) and the feedback inhibition of enzymes (such as citrate synthase and isocitrate dehydrogenase) by the products of the cycle (such as ATP, NADH, and citrate) .
- The Kreb cycle is one of the oldest metabolic pathways, as it is found in almost all living organisms that use oxygen for respiration . It is thought to



have evolved from an ancient anaerobic pathway that used sulfur instead of oxygen as the final electron acceptor.

### Electron Transport System and Oxidative Phosphorylation

- Electron Transport System (ETS) is a series of proteins and electron carriers located in the inner mitochondrial membrane that transfers electrons donated by the reduced molecules NADH and FADH<sub>2</sub> through a series of electron acceptors, to oxygen .
- Oxidative Phosphorylation (OP) is a process involving a flow of electrons through the ETS, which creates an electrochemical gradient of protons across the inner mitochondrial membrane .
- The electrochemical gradient of protons is used to drive the synthesis of ATP from ADP and Pi by a membrane-bound enzyme called ATP synthase, in a process called chemiosmosis .
- Together, the ETS and chemiosmosis make up OP, which is the final stage of cellular respiration and produces most of the ATP generated from the oxidation of glucose .
- The ETS consists of four major protein complexes (I, II, III, and IV) and two mobile electron carriers (ubiquinone and cytochrome c) that shuttle electrons between the complexes .
- Complex I receives electrons from NADH and transfers them to ubiquinone, while pumping four protons across the membrane .
- Complex II receives electrons from FADH<sub>2</sub> and transfers them to ubiquinone, without pumping protons .
- Ubiquinone delivers electrons to complex III, which transfers them to cytochrome c, while pumping four protons across the membrane .
- Cytochrome c delivers electrons to complex IV, which transfers them to oxygen, the final electron acceptor, while pumping two protons across the membrane .
- Oxygen combines with protons to form water as a byproduct of OP .
- The ETS creates a proton gradient that is higher in the intermembrane space than in the matrix of the mitochondria .
- The proton gradient represents a potential energy that can be harnessed by ATP synthase to make ATP .
- ATP synthase is a complex enzyme that has two parts: F<sub>0</sub> and F<sub>1</sub> .
- F<sub>0</sub> is embedded in the membrane and acts as a proton channel that allows protons to flow back into the matrix .
- F<sub>1</sub> is attached to F<sub>0</sub> and protrudes into the matrix, where it catalyzes the phosphorylation of ADP to ATP .
- The flow of protons through F<sub>0</sub> causes F<sub>1</sub> to rotate, which changes its conformation and enables the binding and release of ADP, Pi, and ATP .
- OP can produce up to 34 ATP molecules per glucose molecule, depending on the efficiency of the ETS and the transport of NADH and FADH<sub>2</sub> into the mitochondria .
- OP is regulated by the availability of oxygen, NADH, FADH<sub>2</sub>, ADP, and

Pi, as well as by the feedback inhibition of ATP on some enzymes of the ETS and glycolysis .

- OP is linked to the previous stages of cellular respiration, namely glycolysis, formation of acetyl-CoA, and Krebs cycle, which provide the substrates and cofactors for OP .
- OP is the main source of ATP for most aerobic organisms, and it is essential for the maintenance of cellular functions and homeostasis .

### **ATP and Cellular Respiration**

- ATP (adenosine triphosphate) is the energy-carrying molecule used in cells because it can release energy very quickly.
- ATP is composed of a nitrogenous base (adenine), a sugar (ribose), and three phosphate groups. The bonds between the phosphate groups are high-energy bonds that can be broken to release energy.
- Cellular respiration refers to the breakdown of glucose and other respiratory substrates to make ATP. It involves three main steps: glycolysis, the citric acid cycle, and oxidative phosphorylation .
- Glycolysis is the first step of cellular respiration that occurs in the cytoplasm. It splits one molecule of glucose into two molecules of pyruvate, producing a net of 2 ATP and 2 NADH (a coenzyme that carries electrons) per glucose molecule .
- Pyruvate oxidation is the second step of cellular respiration that occurs in the mitochondrial matrix. It converts each pyruvate molecule into acetyl-CoA (a two-carbon molecule attached to coenzyme A), releasing one molecule of CO<sub>2</sub> and one molecule of NADH per pyruvate molecule .
- The citric acid cycle (also known as the Krebs cycle) is the third step of cellular respiration that occurs in the mitochondrial matrix. It completes the oxidation of glucose by breaking down acetyl-CoA into CO<sub>2</sub> and H<sub>2</sub>O, producing 2 ATP, 6 NADH, and 2 FADH<sub>2</sub> (another coenzyme that carries electrons) per glucose molecule .
- Oxidative phosphorylation is the final step of cellular respiration that occurs in the inner mitochondrial membrane. It uses the electrons carried by NADH and FADH<sub>2</sub> to create a proton gradient across the membrane, which drives the synthesis of ATP by ATP synthase. This process produces about 26-28 ATP per glucose molecule .
- Cellular respiration is regulated by various factors, such as the availability of oxygen, glucose, and other substrates, the demand for ATP, and the feedback inhibition of enzymes .